# THE LEGENDRE EQUATION.

The Legendre equation is given by

$(1-x^2)y''$

$$(1-x^2)\sum_{n=0}^{\infty} n(n-1)a_n x^{n-2} - 2x\sum_{n=1}^{\infty} na_n x^{n-1} + \sum_{n=0}^{\infty} \lambda(\lambda+1)a_n x^n$$

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=2}^{\infty} n(n-1)a_n x^n - \sum_{n=1}^{\infty} 2na_n x^n + \sum_{n=0}^{\infty} \lambda(\lambda+1)a_n$$

Shifting of Sigma notation.

$$\sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n - \sum_{n=0}^{\infty} n(n-1)a_n x^n - \sum_{n=0}^{\infty} 2na_n x^n + \sum_{n=0}^{\infty} \lambda(\lambda+1)a_n x^n =$$

$$(n+2)(n+1)a_{n+2} - n(n-1)a_n - 2na_n + \lambda(\lambda+1)a_n = 0$$

$$a_{n+2} = [n(n-1) + 2n - \lambda(\lambda+1)]a_n$$

$$a_2 = \frac{-\lambda(\lambda+1)a_0}{2}$$

$a_5 = \cancel{60} - [12(\lambda+1)$

When $n=1$

$$a_3 = \frac{2-\lambda(\lambda+1)a_1}{6}$$

$n=2$

$$a_4 = \frac{6-\lambda(\lambda+1)a_2}{12}$$

$$a_4 = \frac{[6-\lambda(\lambda+1)][-\lambda(\lambda+1)a_0]}{24}$$

From Power Series

$$y(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 + a_4x^4 + \cdots + a_nx^n$$

$$y(x) = a_0 + a_1x + -\frac{\lambda(\lambda+1)}{2}a_0x^2 + \frac{2-\lambda(\lambda+1)a_1}{6}x^3$$

$$y(x) = a_0\left[1 - \frac{\lambda(\lambda+1)x^2}{2} + (\lambda(\lambda+1)-6)\lambda(\lambda+1)x^4\right] +$$

$$a_1\left[x + \frac{2-\lambda(\lambda+1)}{6}x^3 + \frac{72-12(\lambda+1)+\lambda(\lambda+1)^2}{120}x^5\right]$$

To find a Polynomial LDE

let $y=1$, $x=1$ $n=\lambda$

$P(x)$ for $n=2$

$$y(x) = a_0\left[1 - \frac{\lambda(\lambda+1)x^2}{2}\right]$$

$$1 = a_0\left[1 - 3x^2\right] \text{ but } x=1$$

$$1 = a_0(-2)$$

$$a_0 = -\frac{1}{2}$$

$$P_2(x) = \frac{1}{2}(3x^2-1)$$

THE RODRIGUES FORMULA

The Rodrigues formula is given by

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n$$

For example: Find $P_2(x)$

$$P_2(x) = \frac{1}{(2^2)(2!)}\frac{d^2}{dx^2}(x^2-1)^2$$

$$= \frac{1}{8}\frac{d^2}{dx^2}(x^4-2x^2+1)$$

$$= \frac{1}{8}\frac{d}{dx}(4x^3-4x)$$

$$= \frac{1}{8}(12x^2-4)$$

$$= \frac{4}{8}(3x^2-1)$$

$$= \frac{1}{2}(3x^2-1)$$

HE BESSEL'S EQUATION.

e Method of Frobenew is Used in finding the ution of Bessel's equation.

Example: $x^2 y_2 + xy_1 + (x^2 - v^2)y = 0$

To find the order of Bessel Differential Equation.

$$x^2 - v^2 = 0$$

$$x^2 = v^2.$$

$$x = \pm v$$

$\therefore$ The B.D.E is of order $v$

Example 2: Determine the order of the following B.D.E

i) $x^2 y_2 + x.y_1 + (x^2 - 1/4)y = 0$

ii) $x^2 y_2 + xy_1 + (x^2 - 1)y = 0$

iii) $xy_2 - 2y_1 + xy = 0$